EEE 215 Artificial Intelligence

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Objectives

This course aims to equip students with the abilities:

- This course provides students an overview of artificial intelligence (AI) principles and approaches.
- Allow students to develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents: Search, Knowledge representation, inference, logic, and learning.

Course Learning Outcomes

Upon the completion of the course, the students are able to:

- Demonstrate broad knowledge of the basic principles of AI.

- Appraise the literature on AI algorithms for given problems.

Programme Learning Outcome

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2				√								

Programme Learning Outcome

PLO1	Engineering Knowledge - Apply knowledge of mathematics, science, electrical & electronic engineering fundamentals and engineering specialisation principles to define and utilize engineering procedures, processes, systems and methodologies.
PLO2	Problem Analysis - Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PLO3	Design/Development of Solutions - Design solutions for complex engineering problems and systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO4	Investigation - Conduct investigation into complex problems using research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PLO5	Modern Tool Usage - Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.
PLO6	The Engineer and Society - Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems (original PLO8)

Program Educational Objectives

1. **Professionalism**-Graduates will be capable of using problem-solving techniques and critical analysis to establish themselves as practicing professionals and demonstrate leadership skills and entrepreneurial spirit in the field of Electrical and Electronics Engineering or related fields.
2. **Continuous Personal Development**-Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.
3. **Societal Engagement**-Graduates will contribute to sustainable development and the well-being of society.

RULES

- Don’t play with your mobile phones
- No eating
- Attendance is **COMPULSORY**
- Absent – medical certificates or related documents are mandatory

Course Outline

Content Outline
1. Introduction to AI and History of AI
2. Problem solving
3. Searching
4. Artificial Neural Networks
5 Machine Learning & Deep Learning
6 Other Neural Networks (Convolutional Neural Network)

Assessment

Assignments	50%
Project Report	50%

Grading System

Grade	Marks	Grade Point	Status
A	85 – 100	4.0	Distinction
A-	80 – 84	3.7	Distinction
B+	75 – 79	3.3	Good
B	70 – 74	3.0	Good
B-	65 – 69	2.7	Good
C+	60 – 64	2.3	Pass
C	55 – 59	2.0	Pass
F	Below 55	0	Fail

Note: you required to pass in the Final Project Report to get overall pass!

Attendance

- You are required to attend the class for more than **80%** of the total credit hours
- If not, you will be **barred** from the final examination/ affect your coursework marks

Contacts / Consultation Hours

- Email : shenyuong.wong@xmu.edu.my
- Room : A1-436
- Phone : 03-87055068

Consultation hour:

- Tuesday (1-3pm)
- Wednesday (10am-12pm)

(Please make appointment with me at least 1 day before).

References

- Stuart Russell. Artificial Intelligence: A Modern Approach (International Edition), 2020.